

From Bragg Rods to Detector Intensity

Calculation sequence

The calculation follows five physical steps:

1. calculate the continuous scattering intensity along every reciprocal rod;
2. rotate the off-specular rods around the film normal to represent random in-plane orientation;
3. broaden the scattering with the crystallite-tilt distribution;
4. retain the reciprocal-space points that satisfy elastic scattering;
5. project the selected scattering through refraction and detector geometry, then add the weighted incident-beam contributions.

Throughout, α denotes crystallite tilt, β crystallite azimuth, and ϕ the outgoing-ray azimuth in the caked image. The sections below develop these steps in order and track the intensity normalization at each change of coordinates.

1 Finite stacking produces continuous rod intensity

The $(10L)$ rod is used to introduce the scattering intensity along one reciprocal rod. The same calculation is then applied to every rod allowed by the experimental geometry. For one in-plane Miller pair (h, k) , the reciprocal vector is

$$\mathbf{Q}_{hk}^0(L) = h\mathbf{a}^* + k\mathbf{b}^* + L\mathbf{c}^*. \quad (1)$$

The coordinate L is measured in the crystallite frame. The complex scattering amplitude of one quintuple layer in orientation σ is

$$F_\sigma(h, k, L; \lambda) = \sum_a o_a f_a(|\mathbf{Q}|, \lambda) \exp[-W_a(\mathbf{Q})] \exp[i\mathbf{Q} \cdot \mathbf{r}_{a,\sigma}]. \quad (2)$$

Here a labels an occupied CIF site, o_a is its occupancy, and $f_a = f_{0,a}(|\mathbf{Q}|/4\pi) + f'_a(\lambda) + if''_a(\lambda)$ is the wavelength-dependent atomic scattering factor in electron units. The displacement factor is written with $W_a(\mathbf{Q}) = \frac{1}{2}\mathbf{Q}^\top \mathbf{U}_a \mathbf{Q}$; for an isotropic site this becomes $\frac{1}{2}|\mathbf{Q}|^2 U_{\text{iso},a}$. The vector $\mathbf{r}_{a,\sigma}$ is the Cartesian position of site a in quintuple-layer orientation σ , and the phase convention is $\exp(+i\mathbf{Q} \cdot \mathbf{r})$.

In the limit of perfect stacking and infinite coherence, interference between repeated layers places intensity at the allowed reciprocal-lattice positions,

$$I_{hk}^\infty(L) = r_e^2 \sum_{\ell \in \mathcal{A}_{hk}} |F_{hk\ell}|^2 \delta(L - \ell), \quad (3)$$

where $\mathcal{A}_{hk}$ contains the allowed integer orders. The three-quintuple-layer repeat of the selected Bi_2Se_3 stacking sequence gives the strong series $L = 3n$.

For a finite ordered stack of N quintuple layers, the coherent amplitude is

$$A_N^{(0)}(h, k, L) = F_+(h, k, L) \sum_{n=0}^{N-1} \exp\left(\frac{2\pi i n L}{3}\right), \quad (4)$$

and the resulting finite-stack scattering intensity is

$$S_{hk}^{(0,N)}(L) = r_e^2 |F_+(h, k, L)|^2 \left| \frac{\sin(N\pi L/3)}{\sin(\pi L/3)} \right|^2. \quad (5)$$

The quantity $S_{hk}(L)$ will be called the rod intensity below; in the present normalization it has units of $\text{\AA}^2$. At $L = 3n$, the interference factor approaches N^2 . A finite stack therefore gives the peaks a width of order $1/N$ along the crystallographic L coordinate and produces the finite-size fringes shown in Fig. 1(b).

The stacking model also allows changes in layer registry and orientation. For one possible layer sequence $\mathcal{P}$,

$$A_N(h, k, L; \mathcal{P}) = \sum_{n=0}^{N-1} F_{\sigma_n}(h, k, L; \lambda) \omega_{hk}^{r_n} \exp\left(\frac{2\pi i n L}{3}\right), \quad \omega_{hk} = \exp\left[\frac{2\pi i (h + 2k)}{3}\right]. \quad (6)$$

The probabilities and coherent amplitudes of all allowed layer sequences are combined analytically through the complete stack. The disorder-averaged rod intensity is

$$S_{hk}(L; \lambda) = r_e^2 \left\langle |A_N(h, k, L; \mathcal{P})|^2 \right\rangle_{\mathcal{P}}. \quad (7)$$

This ensemble average is evaluated directly at every crystallographic L required by the reciprocal-space or detector calculation. The physical progression is

$$\underbrace{\{\ell\}_{\text{Bragg}}}_{\text{infinite ordered crystal}} \longrightarrow \underbrace{S_{hk}^{(0,N)}(L)}_{\text{finite ordered stack}} \longrightarrow \underbrace{S_{hk}(L)}_{\text{stacking model}}. \quad (8)$$

The dimensional distance along the rod is

$$u = |\mathbf{c}^*|L, \quad \mathbf{Q}_{hk}^0(u) = \mathbf{q}_{hk} + u\hat{\mathbf{c}}^*, \quad \mathbf{q}_{hk} = h\mathbf{a}^* + k\mathbf{b}^*, \quad (9)$$

where $\hat{\mathbf{c}}^* = \mathbf{c}^*/|\mathbf{c}^*|$.

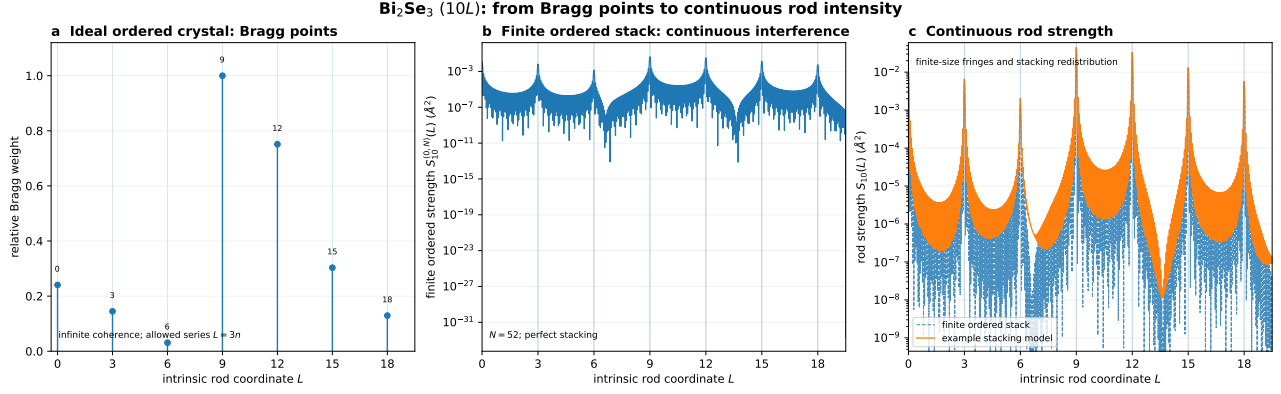


Figure 1: Construction of the Bi₂Se₃ (10L) rod strength. (a) The infinite ordered crystal gives Bragg points at the allowed $L = 3n$ positions. The stems represent delta functions with relative crystallographic weights. (b) A finite ordered stack with $N = 52$ gives finite peak widths and finite-size fringes. (c) The stacking model redistributes intensity within that continuous profile.

2 Elastic reach defines the reciprocal-rod set

A rod can contribute only if an Ewald sphere can reach it for at least one incident wavelength. The shortest distance between the rod and the reciprocal origin is

$$Q_{R,hk} = |\mathbf{q}_{hk,\perp}|, \quad \mathbf{q}_{hk,\perp} = \mathbf{q}_{hk} - (\mathbf{q}_{hk} \cdot \hat{\mathbf{c}}^*) \hat{\mathbf{c}}^*. \quad (10)$$

Crystallite rotation preserves this distance.

An Ewald sphere of radius k , whose center is also a distance k from the reciprocal origin, can reach the rod when

$$Q_{R,hk} \leq 2k. \quad (11)$$

The shortest sampled wavelength has the largest wavevector,

$$k_{\max} = \frac{2\pi}{\lambda_{\min}}. \quad (12)$$

Every integer pair (h, k) satisfying

$$Q_{R,hk} \leq 2k_{\max} \quad (13)$$

is included in the complete rod set. For the Bi₂Se₃ source distribution used here, $\lambda_{\min} = 1.506747 \text{ \AA}$ and $2k_{\max} = 8.340065 \text{ \AA}^{-1}$. The complete rod set contains the $(00L)$ rod and 84 nonzero (h, k) rods.

Each weighted incident-beam component applies the same reach condition using its own wavelength,

$$k_s = \frac{2\pi}{\lambda_s}, \quad Q_{R,hk} \leq 2k_s. \quad (14)$$

Thus the rods available for scattering depend slightly on wavelength. Of the 1000 incident-beam components, 991 can reach all 85 rods. The nine longest-wavelength components cannot reach the outer $r = 21$ family and use 73 rods.

For the hexagonal in-plane lattice, rods with the same

$$r = h^2 + hk + k^2 \quad (15)$$

share the same powder-cylinder radius,

$$Q_R(r) = Q_R(1)\sqrt{r}, \quad Q_R(1) = 1.751194 \text{ \AA}^{-1}. \quad (16)$$

The value r labels a radial family. The number of rods is the number of distinct integer pairs (h, k) at that radius.

Table 1: Nonzero radial families in the Bi_2Se_3 rod set. The final column is the number of distinct (h, k) rods at the listed radius.

r	$Q_R(r) \text{ (\AA}^{-1}\text{)}$	number of rods
1	1.751	6
3	3.033	6
4	3.502	6
7	4.633	12
9	5.254	6
12	6.066	6
13	6.314	12
16	7.005	6
19	7.633	12
21	8.025	12

The factor P_{hk} is the relative population assigned to one (h, k) rod. It allows unequal phase fractions or crystallographic-domain populations. Here $P_{hk} = 1$ for every rod. For a compact figure, the rods at one radius are added as

$$S_{r,s}^\Sigma(L) = \sum_{\substack{(h,k) \text{ available at } \lambda_s \\ h^2+hk+k^2=r}} P_{hk} S_{hk}(L; \lambda_s). \quad (17)$$

This family sum is used only for display. The scattering calculation evaluates every (h, k) rod separately.

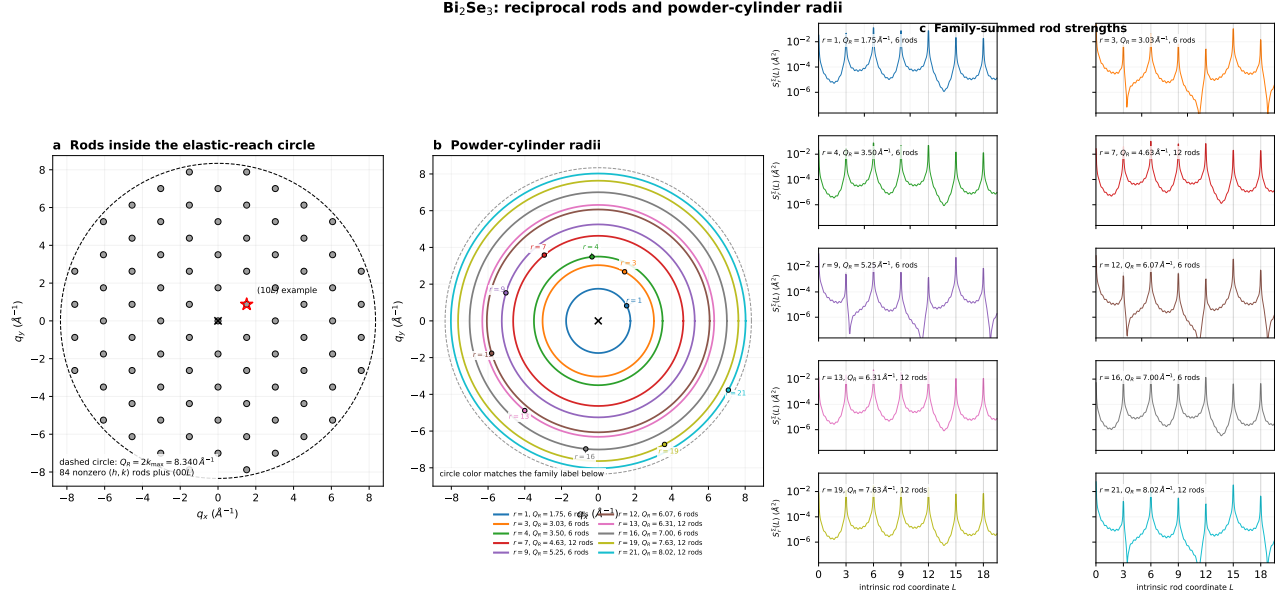


Figure 2: Reciprocal rods and their powder-cylinder radii. (a) Every integer rod inside the elastic-reach circle is included. The red star marks (10L) and the black cross marks (00L). (b) The ten powder-cylinder radii are color coded. Each ring is marked directly by r , and the matching legend gives Q_R and the number of rods. (c) Each small panel gives the family-summed strength $S_r^\Sigma(L)$ for one discrete r .

3 Random in-plane orientation forms normalized powder cylinders

A two-dimensional powder contains crystallites with every in-plane azimuth. The rotation $\mathbf{C}$ places the mean crystallite orientation in the sample frame and aligns the reciprocal stacking direction with the measured film normal,

$$\mathbf{C}\hat{\mathbf{c}}^* = \hat{\mathbf{n}}_{\text{film}}. \quad (18)$$

At zero mosaic tilt, rotation through crystallite azimuth β gives

$$\mathbf{Q}_{hk}^{\text{cyl}}(\beta, u) = \mathbf{C}R_{\hat{\mathbf{c}}^*}(\beta)\mathbf{Q}_{hk}^0(u), \quad 0 \leq \beta < 2\pi. \quad (19)$$

For fixed u , this traces a ring of radius $Q_{R,hk}$. Continuous u , or continuous L , stacks those rings into a cylinder.

A small rod segment du carries

$$dI_{hk,s}^{\text{rod}} = P_{hk}S_{hk}(L; \lambda_s) du. \quad (20)$$

Uniform crystallite azimuth assigns the fraction $d\beta/(2\pi)$ to an angular interval $d\beta$,

$$dI_{hk,s}^{\text{cyl}} = dI_{hk,s}^{\text{rod}} \frac{d\beta}{2\pi}. \quad (21)$$

The corresponding cylinder area is

$$dA_{\text{cyl},hk} = Q_{R,hk} d\beta du. \quad (22)$$

The intensity per reciprocal-space cylinder area is therefore

$$\rho_{\text{cyl},hk,s}(\beta, u) = \frac{P_{hk}S_{hk}(L; \lambda_s)}{2\pi Q_{R,hk}}. \quad (23)$$

Integrating around the circumference recovers the original rod segment,

$$\int_0^{2\pi} \rho_{\text{cyl},hk,s}(\beta, u) Q_{R,hk} d\beta = P_{hk}S_{hk}(L; \lambda_s). \quad (24)$$

The factor $1/(2\pi Q_R)$ spreads the scattering in crystallite azimuth while preserving the intensity at each crystallographic L . Figure 3(a) identifies the ten discrete radial-family cylinders. The stepped display heights expose the nested surfaces, while every physical cylinder retains its full continuous L range. Panels (b) and (c) show the family densities and the numerical circumference integration.

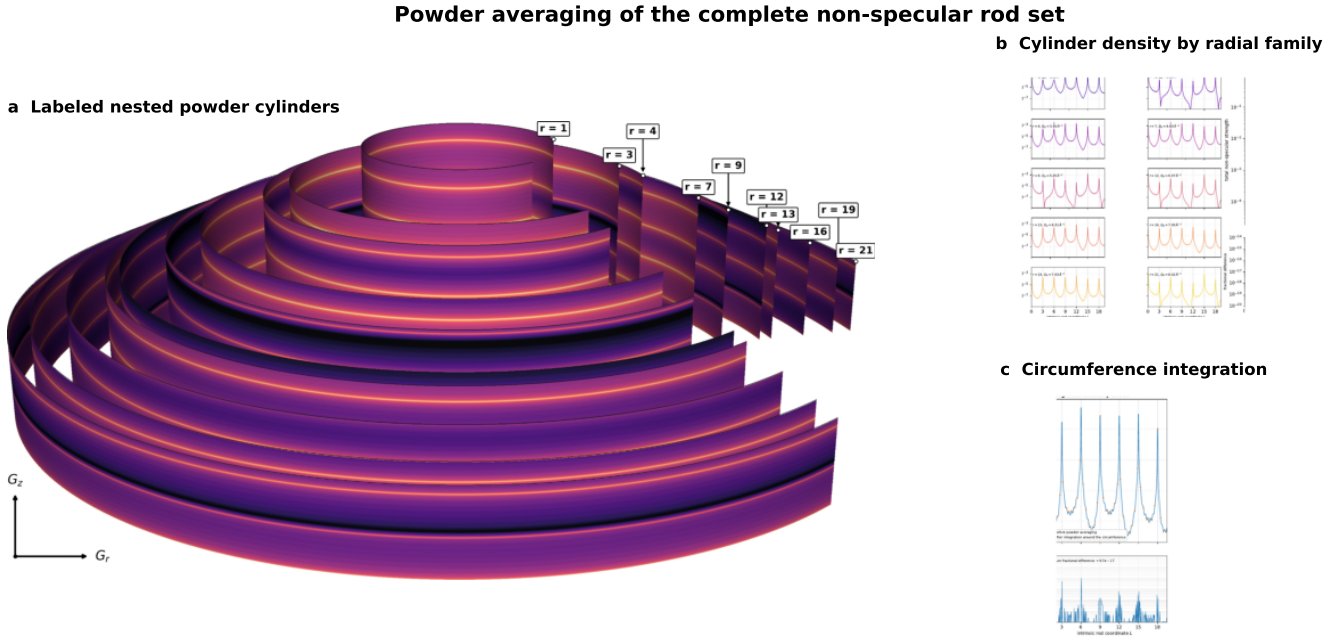


Figure 3: Powder averaging of the complete non-specular rod set. (a) The labeled cutaway identifies the cylinder associated with every discrete radial family r ; the stepped heights are used only to reveal the nested surfaces. (b) The cylinder density $\rho_{\text{cyl},r}(L)$ for each family. (c) Circumference integration reproduces the summed rod intensity to numerical precision.

4 Mosaicity spreads the cylinders into reciprocal-space volume

Mosaicity rotates the complete crystallite, including the rod direction and its in-plane offset. Let α be the tilt magnitude and β the azimuth of that tilt. If $\hat{\mathbf{t}}$ is a reference in-plane tilt axis, then

$$Q_{hk}(\alpha, \beta, u) = C R_{\hat{\mathbf{c}}^*}(\beta) R_{\hat{\mathbf{t}}}(\alpha) Q_{hk}^0(u). \quad (25)$$

The innermost rotation tilts the crystallite by α . The second rotation sets the azimuth β of that tilt around the mean stacking direction. At $\alpha = 0$, the expression reduces to the powder cylinder.

The default Bi_2Se_3 calculation uses a wrapped Gaussian tilt profile with width $\sigma_\alpha = 1^\circ$. Because tilt is an angular variable, the profile is repeated every 2π so that probability is continuous across the angular boundary. The symmetric profile is then folded onto the nonnegative tilt magnitude $0 \leq \alpha \leq \pi$. The orientation model can also include a separately weighted Lorentzian tail, but its weight is zero in the default calculation shown here.

Let $p_\alpha(\alpha)$ denote the resulting normalized tilt probability. Uniform crystallite azimuth gives

$$p_m(\alpha, \beta) = \frac{p_\alpha(\alpha)}{2\pi}, \quad \int_0^\pi \int_0^{2\pi} p_m(\alpha, \beta) d\beta d\alpha = 1. \quad (26)$$

For one rod in incident-beam component s , the intensity carried by $d\alpha d\beta du$ is

$$dI_{hk,s}^{\text{pre}} = p_m(\alpha, \beta) P_{hk} S_{hk}(L; \lambda_s) d\alpha d\beta du. \quad (27)$$

Because p_m is normalized, integration over crystallite orientation recovers the original rod intensity.

The reciprocal-space volume generated by changes in α , β , and u is

$$dV_Q = J_{\text{vol},hk} d\alpha d\beta du, \quad J_{\text{vol},hk} = \left| \det \frac{\partial \mathbf{Q}_{hk}}{\partial(\alpha, \beta, u)} \right|. \quad (28)$$

When a broad range of crystallite orientations is compressed into a small reciprocal-space volume, the intensity per unit volume increases. Adding every rod and crystallite orientation that reaches the same reciprocal-space point gives

$$\rho_{Q,s}^{\neq 0}(\mathbf{Q}) = \sum_{\substack{\text{reachable } (h,k) \neq (0,0) \\ \text{scattering solutions at } \mathbf{Q}}} \frac{p_m(\alpha, \beta) P_{hk} S_{hk}(L; \lambda_s)}{J_{\text{vol},hk}}. \quad (29)$$

The intrinsic L in $S_{hk}(L)$ remains the crystallite-frame rod coordinate. After rotation,

$$Q_z = \mathbf{Q} \cdot \hat{\mathbf{n}}_{\text{film}} \quad (30)$$

contains contributions from both the in-plane reciprocal vector and $L\mathbf{c}^*$. Thus $Q_z c / (2\pi)$ is not generally equal to intrinsic L for a tilted crystallite.

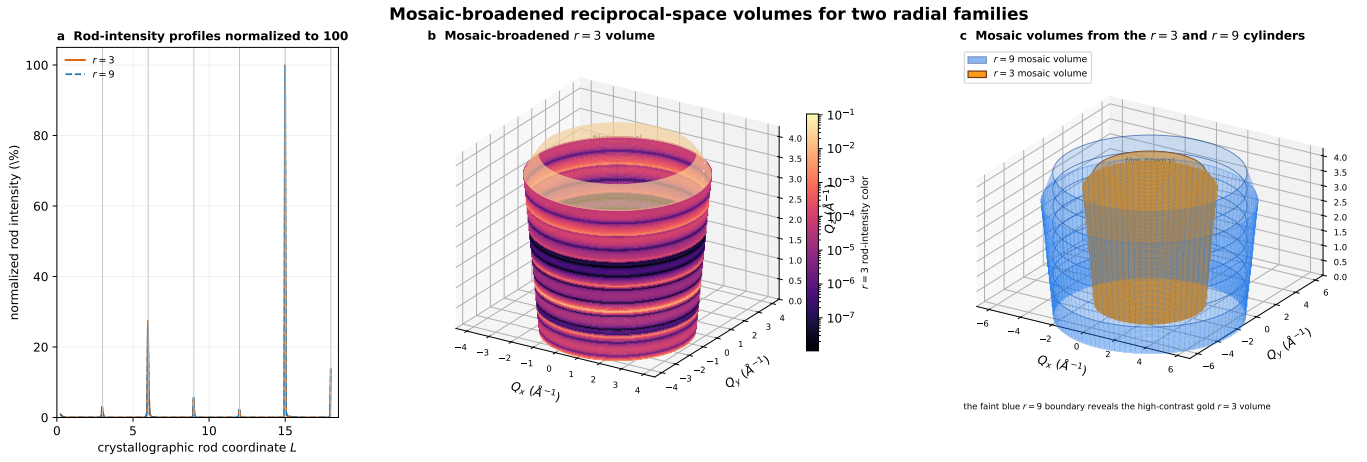


Figure 4: Mosaic broadening of selected off-specular radial families. (a) The $r = 3$ and $r = 9$ rod-intensity profiles are shown on a linear scale, with each profile normalized independently to a maximum of 100 for comparison of their L dependence. (b) The boundary of the reciprocal-space volume formed from the $r = 3$ powder cylinder by the full orientation range $0 \leq \alpha \leq 10^\circ$ and $0 \leq \beta < 2\pi$; the interior contains all smaller tilt magnitudes. The surface color follows the $r = 3$ rod intensity along L . (c) Applying the same mosaic construction to the $r = 3$ and $r = 9$ cylinders produces two nested reciprocal-space volumes. The outer $r = 9$ boundary is rendered as a faint blue shell, while the inner $r = 3$ volume is shown in high-contrast gold so that both families remain visible. The colors identify the families and do not compare their absolute intensities.

5 The $(00L)$ family remains axial until mosaic tilt

The $r = 0$ family contains only the $(00L)$ rod,

$$Q_{00}^0(L) = Lc^*, \quad Q_R = 0. \quad (31)$$

The same ideal-to-finite construction produces a continuous $S_{00}(L)$. The lateral registry phase is unity for $h = k = 0$, so the finite ordered and example stacking-model profiles nearly coincide in the present registry-transition model.

At zero tilt, changing β leaves the $00L$ vector on the specular axis. Once $\alpha > 0$, a fixed crystallographic L moves to

$$Q_R = |Lc^*| \sin \alpha, \quad Q_z = |Lc^*| \cos \alpha, \quad (32)$$

while β distributes the tilted vector around the film normal. Figure 5(c) shows the full $0 \leq \alpha \leq 10^\circ$ broadening of the $L = 9$ and $L = 18$ points. Figure 5(d) integrates continuous L over the same tilt range to give the specular reciprocal-space volume. Figure 5(e) shows the closed boundary of that full volume: the side corresponds to $\alpha = 10^\circ$ and the top cap contains all tilts from 0 to 10° at the largest displayed L .

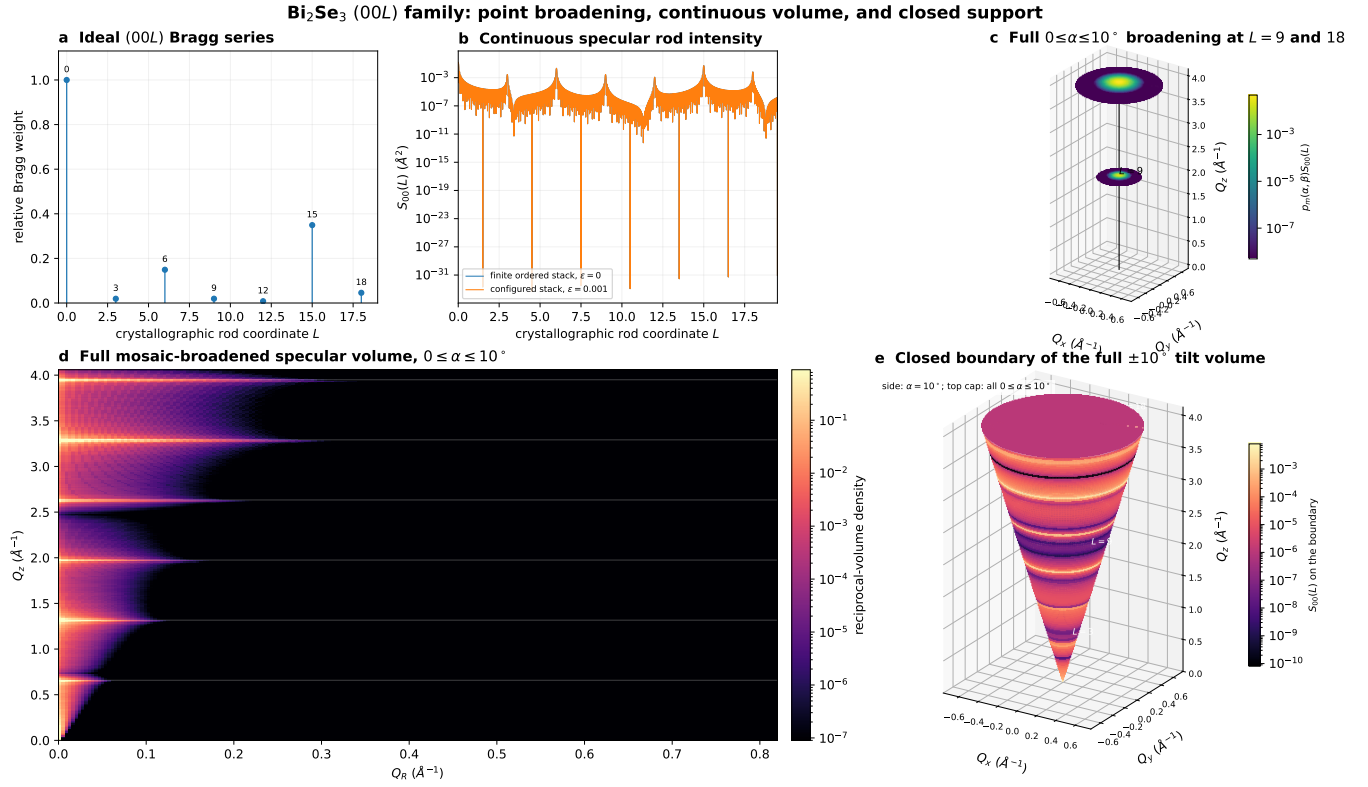


Figure 5: Construction of the Bi₂Se₃ (00L) family. (a) The infinite ordered limit gives the $L = 3n$ Bragg series. (b) The finite ordered and stacking-model profiles nearly overlap for the present $r = 0$ registry phase. (c) The complete $0 \leq \alpha \leq 10^\circ$ orientation range broadens the individual $L = 9$ and $L = 18$ axis points into finite-radius surfaces. (d) Continuous L and the full tilt range produce the mosaic-broadened specular volume. (e) The boundary of that volume is a closed cone: its side is the $\alpha = 10^\circ$ surface and its top cap contains the full range $0 \leq \alpha \leq 10^\circ$ at the largest displayed L .

6 Elastic scattering restricts the reciprocal density to the Ewald sphere

6.1 Off-specular rods form detector-visible Ewald patches

For one incident-beam component s , the internal incident wavevector is $\mathbf{k}_{i,s}$ and

$$K_s = |\mathbf{k}_{i,s}|. \quad (33)$$

Elastic scattering inside the film requires

$$|\mathbf{k}_{f,s}| = K_s, \quad \mathbf{Q} = \mathbf{k}_{f,s} - \mathbf{k}_{i,s}, \quad (34)$$

so the allowed reciprocal vectors lie on

$$|\mathbf{k}_{i,s} + \mathbf{Q}| = K_s. \quad (35)$$

This Ewald sphere has radius K_s , center $-\mathbf{k}_{i,s}$, and the reciprocal origin on its surface.

The preceding reciprocal-space construction supplies a continuous density before this elastic restriction. The Ewald condition retains only the rod and crystallite orientations that intersect the sphere for the selected incident-beam component. The following pages first describe this restriction with the analytic rod–sphere geometry and then show the equivalent detector-coordinate evaluation used by the production calculation.

For a fixed off-specular rod and crystallite orientation, the reciprocal rod is a straight line,

$$\mathbf{Q}(u) = \mathbf{q}_0 + u\hat{\mathbf{d}}, \quad (36)$$

where $\mathbf{q}_0$ is the point at $u = 0$ and $\hat{\mathbf{d}}$ is the rotated rod direction. Defining

$$\mathbf{a} = \mathbf{k}_{i,s} + \mathbf{q}_0, \quad \mathbf{a}_\perp = \mathbf{a} - (\mathbf{a} \cdot \hat{\mathbf{d}})\hat{\mathbf{d}}, \quad (37)$$

the algebraic intersections with the internal Ewald sphere are

$$u_j = -\mathbf{a} \cdot \hat{\mathbf{d}} \pm \sqrt{K_s^2 - |\mathbf{a}_\perp|^2}, \quad L_j = \frac{u_j}{|\mathbf{c}^*|}. \quad (38)$$

A negative value under the square root means that the oriented rod misses the sphere, zero gives tangency, and a positive value gives two algebraic intersections.

These roots are used to construct a forward Ewald-patch picture and to validate the branch geometry. They are not evaluated as a separate line-intersection step when the detector image is calculated. In the detector calculation, a detector coordinate fixes the outgoing ray and therefore fixes a reciprocal-space point on the Ewald sphere,

$$(c, r_d) \longrightarrow \mathbf{k}_{f,s}^{\text{air}} \longrightarrow \mathbf{k}_{f,s}^{\text{film}} \longrightarrow \mathbf{Q}_s \longrightarrow \{h, k, j, \alpha, \beta, L\}_{\text{compatible}}. \quad (39)$$

The inverse reciprocal map then returns every rod, crystallite orientation, branch, and crystallographic L that contributes to that detector point. Figure 6 shows the complete intrinsic Ewald intensity for the nominal 5° incidence state. Figure 7 then isolates the subset that reaches the active detector.

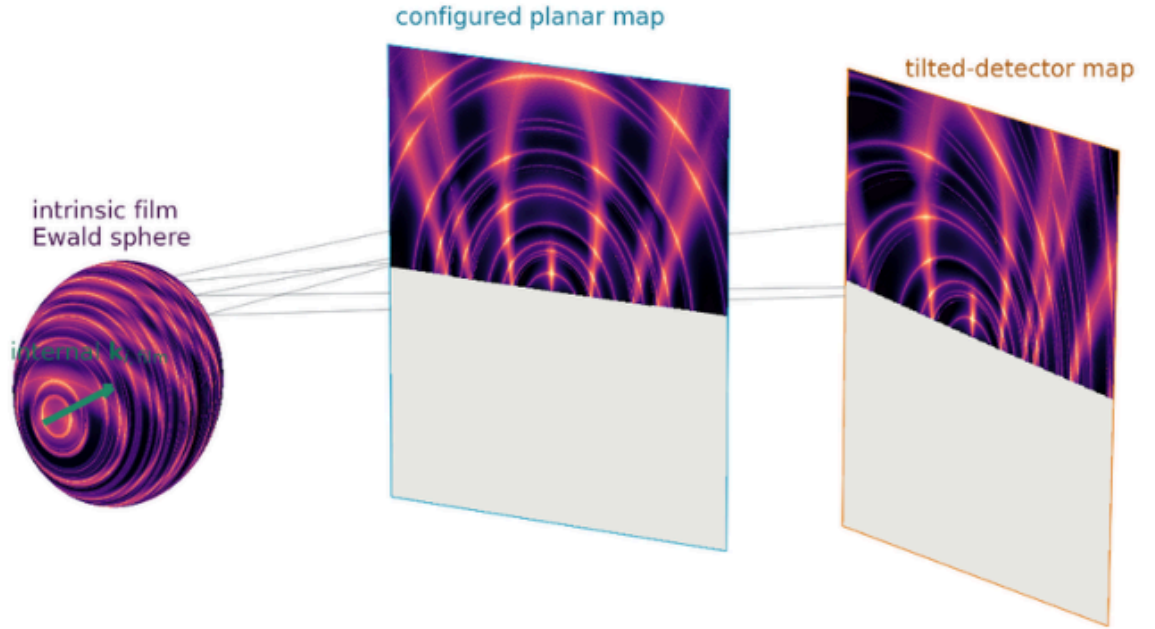
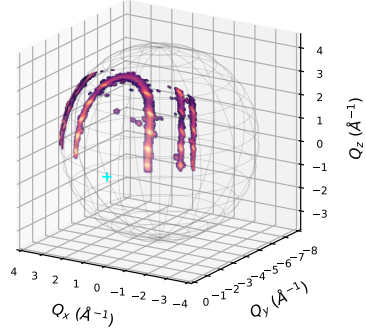


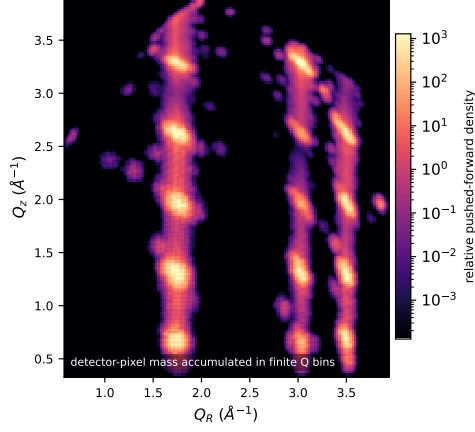
Figure 6: Complete intrinsic Ewald intensity and detector mappings for the nominal Bi_2Se_3 incidence angle of 5° . The full internal-film Ewald sphere is shown at left, followed by the configured planar map and the tilted-detector map after exit refraction. Figure 7 isolates the active-detector subset.

Off-specular detector-visible Ewald support, nominal incidence 5°

a Ewald points: detector-side view



b Area-binned (Q_R, Q_z) density



c Detector density

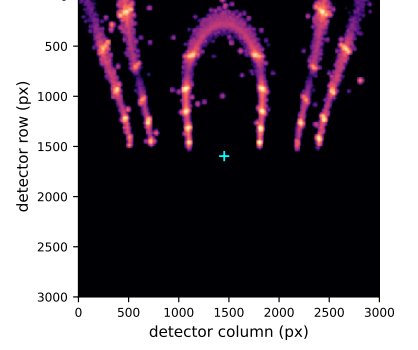


Figure 7: Detector-visible off-specular Ewald selection for the nominal Bi_2Se_3 incidence angle of 5° . (a) The view is rotated toward the detector side so the detector-visible reciprocal-space points lie on the near side of the sphere; the cyan cross marks $\mathbf{Q} = 0$. (b) The detector-pixel intensity is accumulated in finite (Q_R, Q_z) bins and divided by the bin area. This area-binned pushforward removes the visual gaps and artificial under-representation near smaller $|\mathbf{Q}|$ that occur when irregularly spaced mapped points are plotted individually. (c) The detector density that supplies panels (a) and (b). This figure isolates the detector-visible subset of the complete sphere shown in Fig. 6.

6.2 The specular family retains the nonzero kinematic branch

The nonzero kinematic branch is treated separately because the $00L$ rod passes through the reciprocal origin. For one crystallite orientation, write the rod as

$$\mathbf{Q}(u) = u\hat{\mathbf{d}}. \quad (40)$$

The elastic condition is

$$|\mathbf{k}_i + u\hat{\mathbf{d}}|^2 = |\mathbf{k}_i|^2. \quad (41)$$

Expanding the left-hand side gives

$$u(2\mathbf{k}_i \cdot \hat{\mathbf{d}} + u) = 0. \quad (42)$$

The first solution, $u = 0$, is the direct-beam point $\mathbf{Q} = 0$. The second solution is the nonzero kinematic $00L$ branch,

$$u_{00}^{\text{kin}} = -2\mathbf{k}_i \cdot \hat{\mathbf{d}}. \quad (43)$$

If the rod direction is represented by a non-unit vector $\mathbf{d}$, the same derivation gives

$$u_{00}^{\text{kin}} = -2 \frac{\mathbf{k}_i \cdot \mathbf{d}}{\mathbf{d} \cdot \mathbf{d}}. \quad (44)$$

This derivation explains why the nonzero solution is used: the $u = 0$ root identifies the direct beam, while the second root is the kinematic scattering point on the $00L$ rod.

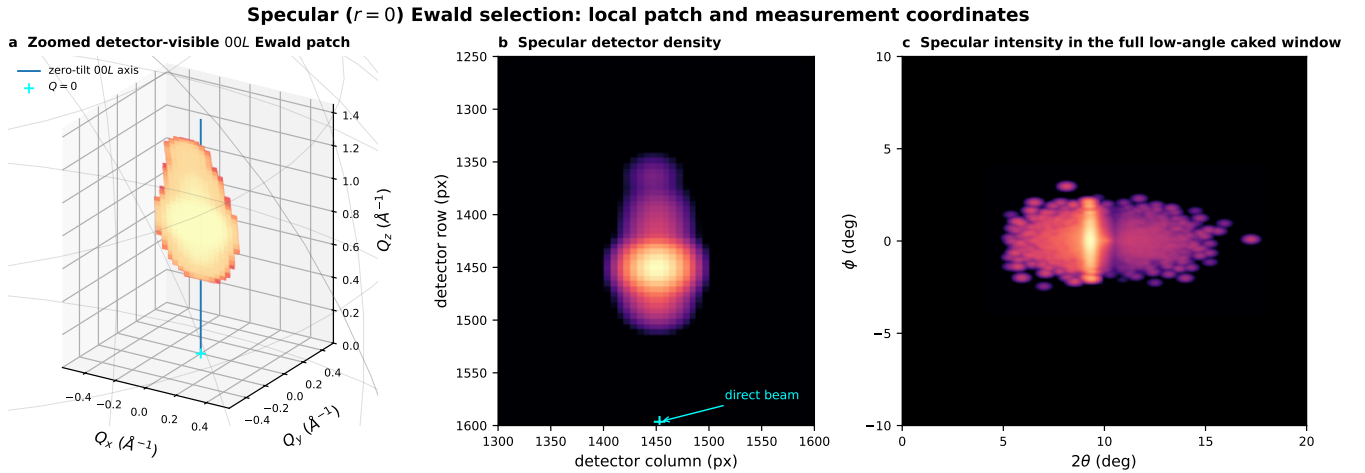


Figure 8: Specular ($r = 0$) Ewald selection for the nominal Bi_2Se_3 incidence angle of 5° . (a) A local view centered on the detector-visible $00L$ Ewald patch, with the zero-tilt $00L$ axis and direct-beam point $\mathbf{Q} = 0$ retained for reference. The zoom emphasizes the region of the Ewald sphere that contributes to the measured specular feature. (b) The corresponding specular detector density; the cyan cross marks the direct-beam coordinate. (c) The same specular intensity in the caked representation over the wider low-angle window $0^\circ \leq 2\theta \leq 20^\circ$ and $-10^\circ \leq \phi \leq 10^\circ$. The calculated feature occupies the central part of this displayed angular region.

6.3 The crossing factor cancels in the physical Ewald-surface density

The Ewald condition restricts a continuous rod coordinate to its regular roots. Let

$$g(u) = |\mathbf{k}_i + \mathbf{Q}(u)| - K. \quad (45)$$

The one-dimensional delta-function identity

$$\int f(u) \delta[g(u)] du = \sum_j \frac{f(u_j)}{|g'(u_j)|}, \quad g(u_j) = 0, \quad (46)$$

shows why a line-sphere crossing factor appears. Since $d\mathbf{Q}/du = \hat{\mathbf{d}}$ and $\mathbf{k}_f = \mathbf{k}_i + \mathbf{Q}(u_j)$,

$$g'(u_j) = \frac{\mathbf{k}_f \cdot \hat{\mathbf{d}}}{K}, \quad J_E = \frac{1}{|g'(u_j)|} = \frac{K}{|\mathbf{k}_f \cdot \hat{\mathbf{d}}|}. \quad (47)$$

Thus J_E is the Jacobian produced by restricting the continuous rod measure to the Ewald sphere. In the forward orientation coordinates, the selected intensity is

$$dI_{hk,j,s}^E = p_m(\alpha, \beta) P_{hk} S_{hk}(L_j; \lambda_s) J_{E,hk,j,s} d\alpha d\beta. \quad (48)$$

The same orientation interval spans the physical Ewald-surface area

$$dA_Q = J_E J_{\text{vol,hk}} d\alpha d\beta, \quad J_{\text{vol,hk}} = \left| \det \frac{\partial \mathbf{Q}}{\partial(\alpha, \beta, u)} \right|. \quad (49)$$

Dividing the selected intensity by this area gives

$$\rho_{Q,\text{surf},s}(\mathbf{Q}) = \sum_{\substack{\text{rods and branches} \\ \text{compatible with } \mathbf{Q}}} \frac{p_m(\alpha, \beta) P_{hk} S_{hk}(L_j; \lambda_s)}{J_{\text{vol,hk}}}. \quad (50)$$

Because $dA_Q = K_s^2 d\Omega_{\text{film}}$, the density per internal-film outgoing solid angle is

$$\rho_{\Omega,\text{film},s} = \sum_{\substack{\text{rods and branches} \\ \text{compatible with } \mathbf{Q}}} \frac{p_m(\alpha, \beta) P_{hk} S_{hk}(L_j; \lambda_s) K_s^2}{J_{\text{vol,hk}}}. \quad (51)$$

7 Detector geometry maps Ewald intensity into detector, angle, and Q coordinates

Each weighted incident-beam component s supplies a wavelength λ_s , incident direction, sample intersection, statistical weight w_s , and internal incident vector $\mathbf{k}_{i,s}^{\text{film}}$. A detector coordinate (c, r_d) identifies a physical detector point and its outgoing air ray. Exit refraction gives the internal outgoing vector $\mathbf{k}_{f,s}^{\text{film}}$, and

$$\mathbf{Q}_s(c, r_d) = \mathbf{k}_{f,s}^{\text{film}}(c, r_d) - \mathbf{k}_{i,s}^{\text{film}}. \quad (52)$$

For this $\mathbf{Q}_s$, the inverse reciprocal map determines every compatible rod, branch, crystallite orientation, and crystallographic L . The rod intensity is evaluated at each recovered L .

For one incident-beam component, the detector intensity density is

$$D_s(c, r_d) = \sum_{\substack{\text{compatible } (h,k),j \\ \alpha, \beta, L}} \frac{p_m(\alpha, \beta) P_{hk} S_{hk}(L; \lambda_s)}{J_{\text{vol,hk}}} J_{Q \leftarrow cr,s} W_{\text{opt},s}, \quad (53)$$

where

$$J_{Q \leftarrow cr, s} = \left| \frac{\partial \mathbf{Q}_s}{\partial c} \times \frac{\partial \mathbf{Q}_s}{\partial r_d} \right| \quad (54)$$

converts detector-coordinate area into internal Ewald-surface area. It contains detector solid angle, detector pose and distance, source-point parallax, and the area change from exit refraction. The optical weight is

$$W_{\text{opt}, s} = |t_{\text{in}, s} t_{\text{out}, s}|^2 A_{\text{depth}, s}. \quad (55)$$

The source-averaged detector density and finite-pixel intensity are

$$D(c, r_d) = \sum_s w_s D_s(c, r_d), \quad I_{pq} = \int_{\text{pixel } pq} D(c, r_d) \, dc \, dr_d. \quad (56)$$

The same detector-visible intensity can be written in caked coordinates $(2\theta, \phi)$,

$$S_{2\theta\phi}(2\theta, \phi) = D(c, r_d) \left| \frac{\partial(c, r_d)}{\partial(2\theta, \phi)} \right|, \quad (57)$$

or in source-resolved internal reciprocal-space coordinates,

$$Q_R = \sqrt{Q_x^2 + Q_y^2}, \quad Q_z = \mathbf{Q} \cdot \hat{\mathbf{n}}_{\text{film}}. \quad (58)$$

The detector-to- $\mathbf{Q}$ transformation is performed separately for each incident-beam component before the intensities are added, because a fixed detector point corresponds to a different $\mathbf{Q}_s$ for different wavelengths and incident directions.

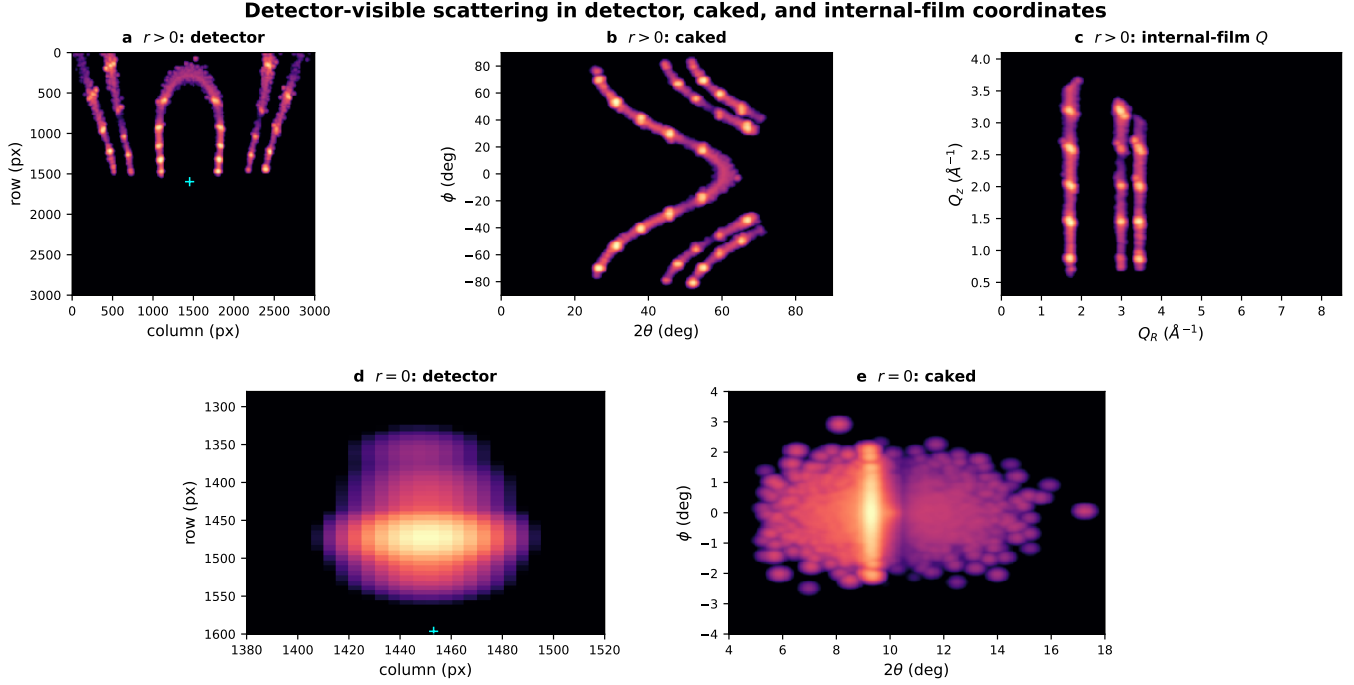


Figure 9: Detector-visible scattering in measurement coordinates. Panels (a)–(c) show the summed off-specular ($r > 0$) intensity in detector, caked, and internal-film (Q_R, Q_z) coordinates. Panels (d) and (e) show the specular ($r = 0$) intensity in detector and caked coordinates. The caked azimuth ϕ describes the outgoing ray, whereas β denotes crystallite azimuth.